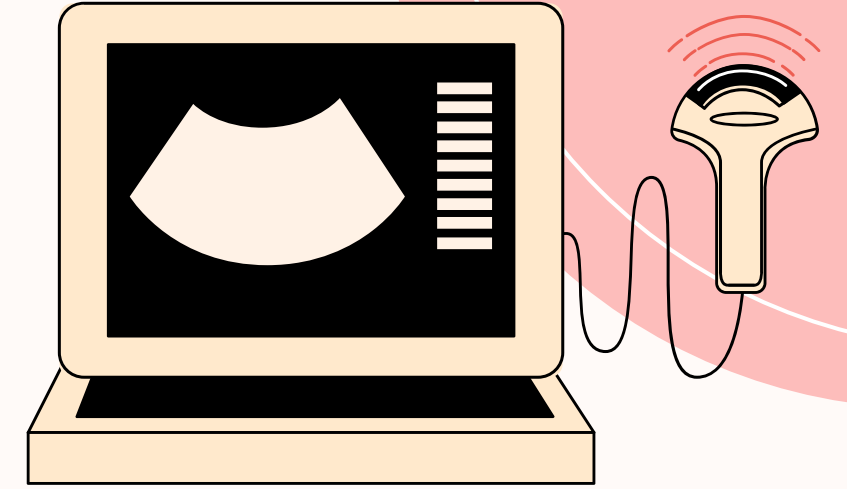


Data-Driven IVF Outcome Analytics & Patient Insights



Company Name: Garbha IVF & Fertility Center

Industry: Healthcare & Fertility Services

Founded: 2018

Location: Hyderabad, India

Employees: 100+

Services: IVF, IUI, fertility consultations, embryo transfer, fertility diagnostics and reproductive health services.



About the Organization



Garbha IVF & Fertility Center is a fictional healthcare organization created for this data analytics case study. The organization specializes in fertility treatments and reproductive healthcare services.

The clinic generates large amounts of patient, treatment and clinical data during every treatment cycle. Analyzing this information can help healthcare professionals monitor outcomes, identify trends and improve treatment strategies.



Business Problem

Current Challenges:

- Patient information and treatment records are collected in large amounts and in multiple systems.
- It is difficult to track and compare IVF treatment outcomes.
- The clinic cannot easily identify the factors that affect treatment success.
- Finding patterns in patient data takes a lot of time.
- Management needs a better way to analyze and visualize treatment data.

Goals And Objectives



- Analyze patient and treatment data.

- Identify patterns that affect IVF success rates.

- Measure important performance indicators (KPIs).

- Compare treatment outcomes across different patient groups.









Key Business Questions



Patient Analysis

- Which age groups have higher IVF success rates?
- Do AMH and AFC levels influence treatment outcomes?



Treatment Analysis

- Which treatment methods are used most frequently?
- Which treatment methods have better success rates?



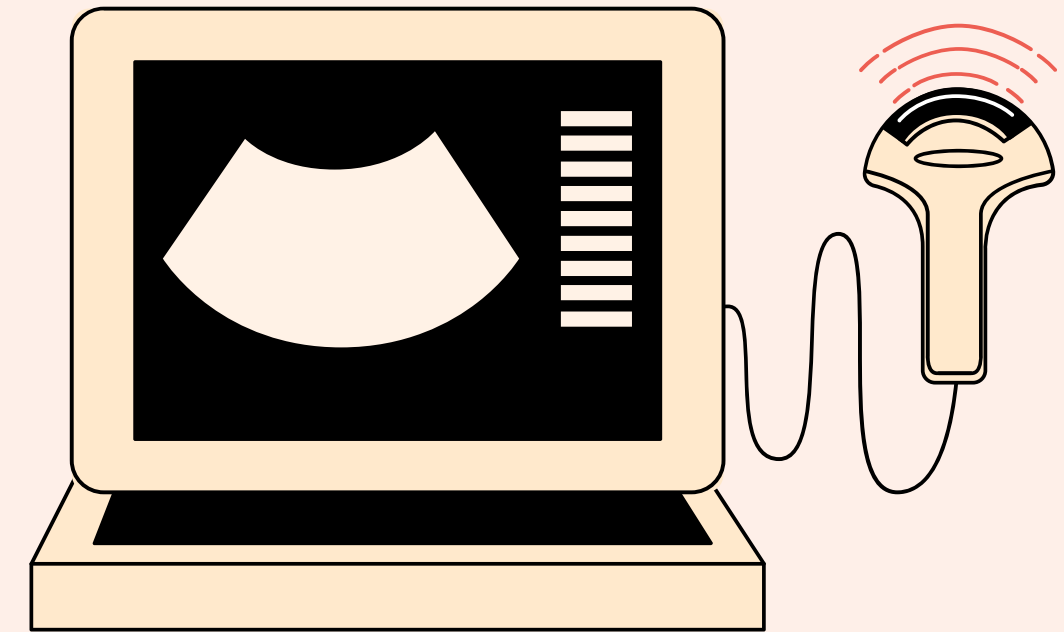
Outcome Analysis

- What is the overall pregnancy success rate?
- What factors are associated with successful treatment outcomes?



Performance Monitoring

- How can treatment data be visualized to support better decision-making?



Dataset Overview



Dataset: IVF Data-Driven Insights – **Synthetic Dataset**

Records: **1,500 IVF treatment cycles**

Variables: **23**

Data Type: **Synthetic / simulated healthcare data**

Data Categories-

 **Patient Profile-** Patient Age, Age Group, Diagnosis, Previous IVF, Smoking Status

 **Clinical & Ovarian Factors-** AMH, AFC Count, Stimulation Days, Endometrial Thickness

 **Treatment & Embryology-** Eggs Retrieved, Mature Eggs, Fertilized Eggs, Embryos Developed, Good Quality Embryos, Fertilization Method, Embryo Stage, Embryo Transfer Count, Embryo Freezing, PGT-A

 **Treatment Outcomes-** Clinical Pregnancy, Live Birth

 **Cost-** Cycle Cost (INR)





Data Dictionary

Patient & Clinical Variables

Field	Description	Type
Cycle_ID	Unique identifier for each IVF cycle	Text
Patient_Age	Age of patient at time of cycle	Numeric
Age_Group	Binned age category	Categorical
Diagnosis	Primary infertility diagnosis	Categorical
Previous_IVF	Whether patient had a prior IVF cycle	Categorical
Smoking_Status	Patient smoking status	Categorical
AMH_ng_mL	Anti-Müllerian Hormone level	Numeric
AFC_Count	Antral Follicle Count	Numeric
Stimulation_Days	Days of ovarian stimulation	Numeric
Eggs_Retrieved	Total eggs collected	Numeric
Mature_Eggs	Eggs mature enough for fertilization	Numeric
Fertilized_Eggs	Eggs successfully fertilized	Numeric

Data Dictionary

(Procedure, Outcome & Cost Variables)

Field	Description	Type
Embryos_Developed	Embryos that developed post-fertilization	Numeric
Good_Quality_Embryos	Embryos meeting quality grading criteria	Numeric
Fertilization_Method	Method used	Categorical
Embryo_Stage	Stage at transfer/freeze	Categorical
Embryo_Transfer_Count	Number of embryos transferred	Numeric
Endometrial_Thickness_mm	Uterine lining thickness	Numeric
Embryo_Freezing	Whether embryos were frozen	Categorical
PGT_A_Performed	Whether genetic testing was done	Categorical
Clinical_Pregnancy	Pregnancy confirmed clinically	Categorical
Live_Birth	Resulted in live birth	Categorical
Cycle_Cost_INR	Total cost of the cycle	Numeric

Stakeholders

Stakeholder	Interest / Stake	What They Want from This Analysis
IVF Clinic Management	Business performance, revenue, growth	Insights on success rates vs. cost to improve profitability
Fertility Specialists / Doctors	Clinical outcomes, treatment protocols	Which factors (age, AMH, stimulation days, method) drive success
Embryologists / Lab Team	Egg/embryo quality and handling	Patterns in fertilization method, embryo stage, and quality outcomes
Patients / Couples	Chances of success, cost transparency	Realistic expectations on success rates and cost planning
Finance / Billing Department	Cost management, pricing strategy	Cost drivers relative to outcomes, to optimize pricing
Data / Analytics Team	Deliver actionable insights	Clean, reliable data and clear recommendations



Data Preparation & Data Cleaning

Data Cleaning Process

- ✓ Data Import- Imported the CSV dataset into Excel, Python, SQL Server, and Power BI.
- ✓ Data Validation- Verified data types., Checked column names, Reviewed data consistency.
- ✓ Missing Value Analysis- Identified missing values, Handled incomplete records.
- ✓ Duplicate Analysis- Checked for duplicate entries, Ensured the uniqueness of treatment-cycle records.
- ✓ Data Transformation- Standardized data for analysis, Prepared the dataset for visualization and reporting.



KPI'S

0.24

Live Birth Rate

1500

Total_Cycles

2.63

Average_AMH

31.71

Average_Age

Key Insights

-  **Age influences IVF outcomes-** The 22–30 age group had the highest live birth rate (29%). The 39–42 age group had the lowest live birth rate (6%).
-  **Unexplained infertility was the most common diagnosis-** It accounted for 301 treatment cycles, the highest among all diagnoses.
-  **ICSI was the preferred fertilization method-** Around 65% of treatment cycles used ICSI.
-  **Treatment outcomes varied across diagnoses-** Success rates differed among different infertility conditions.
-  **Embryo numbers decreased at each IVF stage-** A gradual decline was observed from fertilization to good-quality embryos.

Recommendations

-  Implement age-based treatment strategies.
-  Use diagnosis-specific treatment planning.
-  Monitor key IVF performance indicators regularly.
-  Expand the use of data-driven decision-making.
-  Use interactive dashboards for continuous monitoring.



Thank You

